

Design FMEA - Smartphone High-Voltage Fast-Charge Cell (t6_r3)

Doc No: VBF-T6R3-DFMEA-01

Case: t6_r3

Rev: 01

Date: 2026-08-26

Scales: Severity (S), Occurrence (O), Detection (D): 1-10; RPN = S x O x D. Virtual-design FMEA on the DFN-verified cell.

No	Item / function	Failure mode	Effect	S	Cause	O	Current control (virtual evidence)	D	RPN	Recommended action
1	Anode, 4C charge	Li plating (local anode potential < 0 V)	Capacity loss, dendrite/short risk, thermal hazard	9	Thin negative electrode, high areal current, electrolyte depletion	2	N/P 1.579 + 60 um cathode collapse CC to ~16.2 ms; DFN min anode pot. +0.1050 V (margin 105.0 mV)	3	54	Deep-start 4C DFN run (DVPR row 8); physical reference-electrode test
2	Cell thermal, 4C charge	T_max > 50 C	Electrolyte degradation, safety limit breach	8	Ohmic heat at 18 A, insufficient cooling	2	h = 250 W/m2/K; DFN T_max 318.922 K (margin 4.23 K)	2	32	Calorimetry on physical build; verify h assumption
3	Anode SEI, 100 cycles	SEI thickness > 500 nm	Impedance rise, capacity fade, plating onset	6	EC reduction kinetics at anode	3	k_sei 5e-14 m/s coating + EC-lean 2000 mol/m3; DFN 313.7 nm (margin 186.3 nm)	3	54	Longer aging horizon (500 cycles) simulation
4	Cell voltage, discharge	Plateau < 4.1 V	Energy/UX shortfall vs spec	5	Electrolyte concentration gradients (DFN-level polarization)	2	sigma 2.0 S/m + t+ 0.5; DFN midpoint 4.1551 V (margin 55.1 mV)	2	20	None beyond current design; monitor in physical DV
5	Cell energy density	ED < 950 Wh/L	Requirement miss	6	Inactive mass (CC foils, separator) and volume overhead	2	Thin 60/109 um stack, 209 um total; DFN 983.61 Wh/L (margin +33.61)	1	12	None beyond current design
6	Simulation fidelity	Proxy-model divergence (SPMe-class error) leads to wrong design sign-off	Field failure of a virtually-passed cell	7	Model order reduction, unmodeled physics	4	DFN everywhere (round-9 SPM rejection logged); honest limitation disclosure	2	56	True DFT/MD when real_compute available; physical prototype DV
7	Aging model	Capacity trajectory artifact misread as real	Wrong durability narrative	3	SEI-model lithium-inventory simplification	5	SEI thickness used as durable metric; artifact labeled in datasheet /DVPR	2	30	Full-chemistry aging model or experimental check

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8	Electrolyte properties	sigma / t+ drift from fixed assumed values	Plateau and plating margins erode	5	Concentration/temperature dependence replaced by fixed 2.0 S/m, 0.5	3	EC-lean gradient reduction; margins: plateau +55.1 mV, plating +105.0 mV	3	45	Run DFN with Nym an2008 concentration-dependent conductivity

Notes

- Highest RPN = 56 (model fidelity) and 54 (plating, SEI) - actions tracked in DVPR rows 8-9.
- All 'current control' evidence values are DFN outputs from r10_V27_combo outputs; no physical test data is implied.